

Stratum V1, V2, and DATUM

Mining jobs and block-template control

Agenda

- Stratum V1 jobs and shares

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- Who controls a V1 block template

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- DATUM's gateway model

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- Stratum V2's three protocols

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- DATUM's gateway model
- Stratum V2's three protocols
- Why Braiins favors Stratum V2

Stratum V1

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- **Share:** valid proof of work for the pool, usually not a valid block
- **Coinbase transaction:** first transaction; claims the subsidy and fees and carries the extranonce

Stratum V1 architecture



Jobs go right. Shares go left. The pool supplies the template.

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- Pool validates the shares and pays for the measured work

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- None of them changes the non-coinbase transaction set

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- The pool's node and template builder select the transactions
- A miner can point hash rate elsewhere, but cannot edit the current job
- A few pool operators therefore choose block contents for a large share of Bitcoin's hash rate

V1 limitations

Property	Consequence
Pool-built templates	Transaction selection is concentrated at pools
JSON messages	More bytes and parsing than a binary protocol needs
Several dialects and extensions	Interoperability depends on implementation details
No native authentication or encryption	Jobs and credentials need external protection
Per-device work construction	More coinbase and Merkle-path work across the stack

Template Control

Split construction from accounting

Template construction

Transactions in the candidate block

Pool accounting

Accepted shares and miner payouts

Both protocols let the miner build the template while the pool tracks shares and pays miners.

Common result

- The miner's Bitcoin node supplies the template

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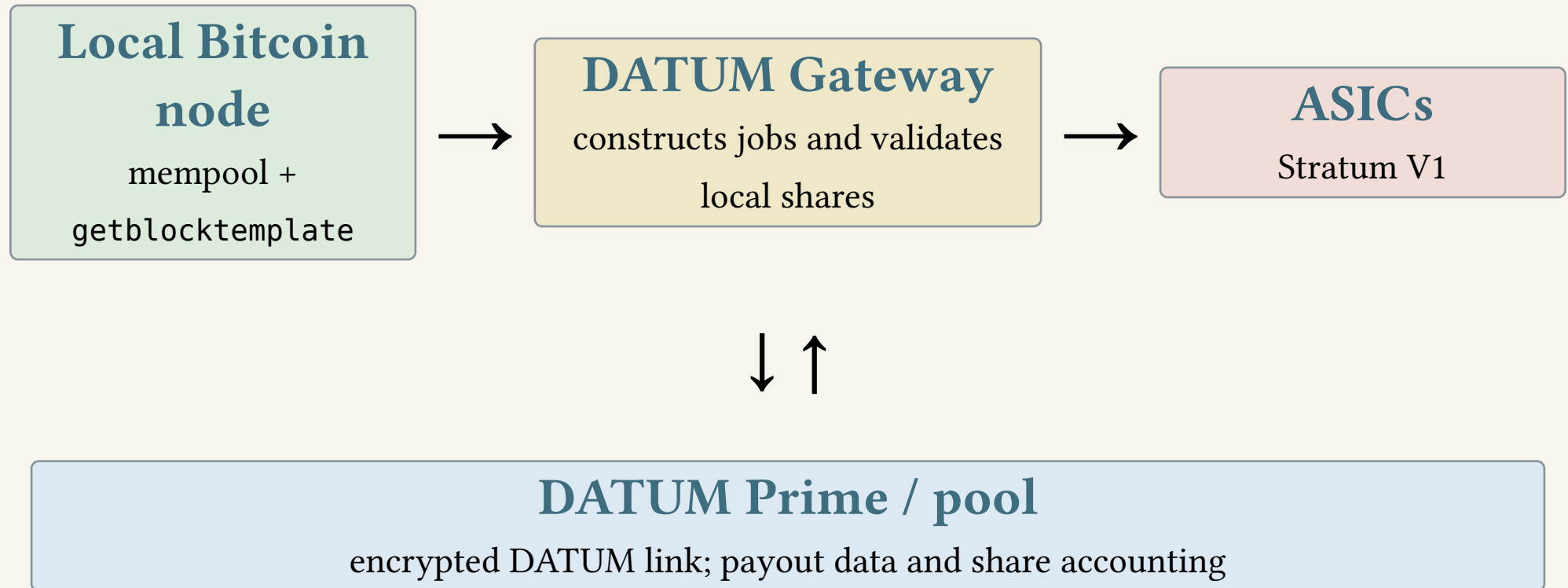
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Common result

- The miner's Bitcoin node supplies the template
- Its mempool and policy determine the transaction set
- The pool still accounts for shares and pays miners
- The miner's node can publish a found block

DATUM

DATUM architecture



How DATUM works

DATUM: Decentralized Alternative Templates for Universal Mining

1. The local node builds a template and returns it through `getblocktemplate`
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5. The Gateway checks shares and forwards qualifying ones to the pool
6. The miner's node publishes a solved block

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- DATUM defines no path for a pool-supplied template
- The pool cannot replace the local template without a configuration change
- This guarantees local control but limits DATUM to that use case

DATUM scope

Changed	Still present
Miner's node selects transactions	Stratum V1 between Gateway and ASICs
Solved blocks submitted locally	Bitcoin RPC and getblocktemplate on the node side
Encrypted Gateway-to-pool link	A custom DATUM pool protocol
Pool accounting without pool work	Pool protocol still described as evolving

Stratum V2

Three protocols

Protocol	Responsibility
Mining Protocol	Jobs, channels, targets, shares, and solutions
Template Distribution Protocol	Templates and transaction data from a full node
Job Declaration Protocol	Miner-created jobs proposed to a pool

The Mining Protocol can run alone. Job Declaration is optional, so SV2 does not automatically decentralize template construction.

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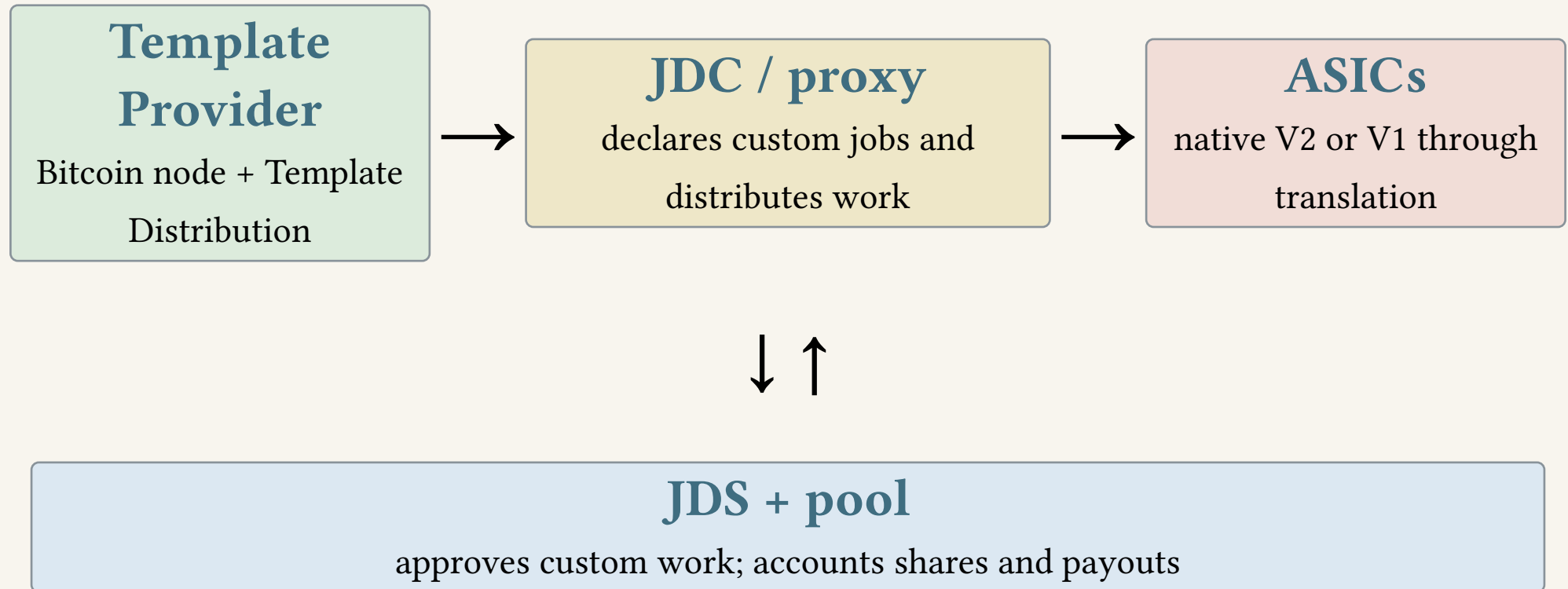
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- **Mining proxy**: distributes jobs and aggregates downstream connections

Stratum V2 architecture



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5. The pool accepts the custom job and agrees to pay its shares
6. The JDC distributes work and forwards shares; the miner's node publishes a found block

Two Job Declaration modes

Property	Full-template	Coinbase-only
Transaction set	Declared by transaction IDs; missing data can be requested	Not revealed to pool or JDS
Fee and validity checks	JDS can verify the proposed template	Pool cannot independently verify the transaction set
Privacy	JDS learns the template	Miner's mempool remains private
Tradeoff	More verification	More withholding / invalid- template risk for pool

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- Future jobs prepared before a new previous-block hash arrives
- Native version rolling and specified protocol extensions

Comparison

SV2 and DATUM

Dimension	Stratum V2	DATUM
Scope	General mining protocol suite	Gateway for local templates and pool accounting
ASIC-facing link	Native V2, or translated V1	V1 today
Node-facing link	Template Distribution Protocol	getblocktemplate over RPC
Template source	Pool jobs or miner custom jobs	Miner's local node only
Template validation	Full-template or coinbase-only mode	Pool-coordinated today; intended to become mostly blind
Specification	Published modular protocol	Open implementation; pool protocol described as evolving
Remote security	Specified Noise/AEAD transport	Encrypted custom DATUM transport

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- Allows pool jobs and V1 translation during migration

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- There is no pool-supplied fallback path

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- This is not a neutral comparison
- DATUM fixes pool-controlled transaction selection
- It also adds another pool protocol while keeping V1 on the ASIC side
- We would rather deploy the SV2 Mining, Template Distribution, and Job Declaration protocols

Comparison

Q&A